Ahmed Hassan Mostafa

Full Stack Developer (React) & AI Automation

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SKILLS

Languages

TypeScript, JavaScript, Python, Dart, C#, Java (Android bridges)

Frontend

React 18/19, Tailwind CSS, Vite, WebGL2

Backend

Node.js, Express, FastAPI, REST & WebSocket APIs, embedded HTTP servers

Databases

SQLite, IndexedDB, Firebase Firestore

AI & Automation

Gemini API, faster-whisper, NLLB-200, Piper TTS, Silero VAD, CUDA, chatbots, WhatsApp/Telegram/Meta webhooks

Desktop & Mobile

Electron, Flutter (Windows/Android), Capacitor (Android/iOS)

Security & Other

AES-256, TOTP 2FA, Ed25519 licence signing, GIS & georeferencing (proj4), DWG/DXF, installers (Inno Setup, NSIS), automated testing, Git

EDUCATION

Pharos University in Alexandria

B.Sc. Computer Engineering — 3rd Year, double-major track

`In progress`

LANGUAGES

Arabic	Native
English	Strong reading & writing, developing speaking

PROFILE

Third-year Computer Engineering student, entirely self-taught, who ships complete products: full-stack web apps with React and Node.js, AI-driven automation, and offline-first desktop and mobile software. 14 projects shipped, including a WebGL2 CAD engine for fibre-network engineers and a real-time voice translator that runs 100% on-device.

PROJECTS

FiberLens — Smart CAD & FTTH Field Engineering Platform

- Offline-first CAD platform that opens and edits DWG/DXF on Android, iOS and the web, with the engineer's live GPS position shown inside the drawing.
- Wrote a WebGL2 renderer that draws 85k entities (~550k segments) in under 1 ms of GPU time; added fibre tracing to the OLT, splitters, georeferencing, BOQ and 10 reports. 38 tests.

`TypeScript` `React` `WebGL2` `Capacitor` `proj4`

`github.com/Ahshika/FiberLens` `Live: fiberlens.pages.dev`

AI Live Translator — Real-time Two-way Voice Translation

- Fully local pipeline (VAD → Whisper → NLLB-200 → Piper) that translates both sides of a Zoom, Teams or Meet call through a virtual mic, in 103 languages.
- C# helper on the Windows process-loopback API captures only the meeting's audio; CUDA with CPU fallback, one-click installer, 85 tests.

`Python` `Whisper` `NLLB-200` `FastAPI` `C#`

`github.com/Ahshika/AI-Live-Translator`

Order Ly — Cafe POS with QR Self-Ordering

- Customers scan the table QR to order and pay from their phone (wallets / InstaPay with a transfer screenshot) and track it live; Flutter app with a self-restarting local server on the cashier PC, Firebase relay, 63 tests + 2 k6 load tests.

`Flutter` `Dart` `SQLite` `Firebase` `QR`

github.com/Ahshika/Order-Ly

FixTrack — Phone Repair Shop & POS System

- One Flutter codebase for Windows and Android: repair workflow (intake → approval → warranty delivery) plus barcode POS, IMEI tracking, used-phone buying and mobile wallets.
- The shop PC runs an embedded HTTP server and SQLite that phones join over Wi-Fi, offline; Firebase QR tracking, Ed25519-signed licences, 104 tests + 5 k6 load tests.

`Flutter` `Dart` `Riverpod` `SQLite` `Firebase`

github.com/Ahshika/FixTrack

NightGold V3 — Password Manager & File Locker

- Offline Windows password manager unlocked by a TOTP code instead of a master password: AES-256 vault, Arabic-capable generator, weak/reused-password audit.
- Quick Lock encrypts any file or folder in place (scrypt + AES-256) and re-locks it on Windows lock, sleep or shutdown.

`Electron` `React 19` `SQLite` `AES-256` `TOTP`

github.com/Ahshika/NightGold-Password-Manager

GeoSnap Pro — GPS Camera & Inspection Reports

- Android inspection camera: live stamps in four templates, real EXIF GPS on every photo, drag-the-pin GPS correction.
- In-app gallery, PDF inspection reports, a native Java bridge, signed and obfuscated releases.

`Capacitor` `JavaScript` `Java` `Leaflet` `jsPDF`

github.com/Ahshika/GeoSnap-Pro-V2

Password Generator V2 — NightGold V2

- Generator with a live strength meter and a vault (search, favourites, categories) unlocked by TOTP paired through a QR code.

`Electron` `React 19` `Tailwind` `SQLite`

github.com/Ahshika/Password-Generator-V2

Password Generator V1 — NightGold V1

- First version: 8–64 character passwords (optional Arabic letters) saved in an AES-256-CBC encrypted SQLite vault.

`Electron` `React` `Vite` `SQLite`

github.com/Ahshika/Password-Generator-V1

Wedding & Event Decor AI Bot

- Live webhook linking Messenger and WhatsApp Cloud API to Gemini; an Egyptian-Arabic chat that qualifies leads.

`Node.js` `Express` `Gemini` `Meta API`

github.com/Ahshika/wedding-decor-bot

Opti Nav — FiberSmart CAD Desktop

- Desktop CAD viewer that led to FiberLens: multi-document tabs, layouts, markup, parser fallback for DWG, DXF, STEP, IGES, STL.

`Electron` `React` `TypeScript` `Zustand`

github.com/Ahshika/Opti-Nav

One Brain

- Daily brain-puzzle game for Android: a shared daily puzzle + an endless Challenge Path that creates its own stages. 8 puzzle types, Java/OOP, AdMob + Play Billing, Arabic/English.

`admob` `android` `arabic` `brain-training` `game`

github.com/Ahshika/One-Brain

Basket

- Supermarket & grocery POS for Windows + Android: scale barcodes, expiry dates (FEFO), multiple cashiers, delivery orders (Flutter, offline-first, LAN server)

`android` `barcode` `dart` `flutter` `grocery`

github.com/Ahshika/Basket

Games

- Games portfolio of Ahmed Hassan — pick a title, read the briefing, download and play.

`android-game` `game-portfolio` `gamepad` `games` `github-pages`

github.com/Ahshika/games

Developers Room

- New project on GitHub.

`TypeScript`

github.com/Ahshika/developers-room